

Interactive Dashboard and Tableau Storytelling

Dataset Link: [Hospital mangement](#)

Students should submit:

Tableau Workbook file (.twb or .twbx)

Steps for each output (brief explanation)

Screenshots of Final Story Points

Short explanation of insights derived from each visualization

Problem Scenario

You are working as a Healthcare Data Analyst.

A hospital management team wants to analyze patient trends, treatment cost, department performance, and regional insights using Tableau Dashboards and Story Points.

Your task is to build interactive visuals and present insights through storytelling.

Q1. Explain the difference between a Tableau Dashboard and a Tableau Story Point. Give two healthcare related use cases.

Ans.

- **Tableau Dashboard:** A collection of several worksheets and objects in a single view. It is designed for **real-time monitoring** and interactivity, allowing users to filter and drill down into data simultaneously.
- **Tableau Story Point:** A sequence of visualizations or dashboards that work together to convey information. It is designed for **narrative storytelling**, guiding the viewer through a specific discovery or logical flow.
- **Healthcare Use Cases:**

Dashboard: A "Live ER Monitor" showing current patient wait times, available beds, and active staff across departments.

Story Point: A "Quarterly Budget Review" presentation showing how treatment costs rose in March, peaked in July, and the specific departments that drove those costs.

Q2. What are actions in tableau ? Explain how filter action can help hospital management analyze patient data.

Ans.

- **Actions:** These are interactive elements that allow you to link different sheets. When you interact with one mark (like clicking a bar), Tableau can perform an action—such as filtering other charts or navigating to a different URL.
- **Filter Action Benefit:** In hospital management, a Filter Action allows an administrator to click on a specific "Hospital Region" in a bar chart. Instantly, all other charts (like Department Performance or Patient Outcomes) update to show data *only* for that region, allowing for localized problem-solving.

Q3.

Ans. Patient Distribution

Insight: This chart identifies which geographical areas contribute the highest volume of patients to the hospital.

Application: Management can use this to determine where to focus marketing efforts or where to establish satellite clinics to reduce travel time for the largest patient clusters.

Q4.

Ans. Department Performance

Insight: This visualization reveals which medical departments are the primary drivers of hospital revenue and resource consumption.

Application: By identifying the highest-cost departments, administrators can perform deeper audits to see if high costs are due to efficient high-volume care or expensive equipment requirements.

Q5.

Ans. Treatment Trend Over Time

Insight: By filtering data from March to September, this line chart highlights seasonal fluctuations in treatment costs and patient intake.

Application: Significant peaks during these months may indicate seasonal health trends, allowing the hospital to plan staffing levels and inventory more accurately for those periods.

Q6.

Ans. Outcome Analysis

Insight: This chart provides a clear ratio of "Recovered" versus "Critical" and "Under Treatment" patients.

Application: It serves as a high-level KPI for hospital effectiveness; a high "Critical" percentage in specific areas may signal a need for improved specialized care or better emergency response protocols.

Q7.

Ans. • **Insight:** This ranked analysis identifies the specific physicians who are managing the most significant portions of the hospital's treatment workload.

• **Application:** This helps in recognizing top performers and identifying potential burnout risks if a small number of doctors are handling a disproportionate number of cases.

Q10. Based on your visuals, answer:

Which department generates the highest treatment cost?

Which region has more critical patients?

What business decisions can hospital management take from these insights?

Ans.

• **Highest Cost Department:** Identified by looking at the tallest bar in the **Department Performance** chart. **Neurology department**

• **Critical Patients by Region:** Found by using the **Outcome Pie Chart** filtered by each region in the dashboard. **West region**

• **Business Decisions:**

Resource Allocation: Move more staff or funding to the department with the highest treatment costs.

Specialized Care: If a specific region has high "Critical" counts, consider building a specialized Intensive Care Unit (ICU) in that area.

Improve treatment quality in regions with many critical patients.